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CREATING KNOWLEDGE CAPITAL: AN ESSENTIAL COMPETITIVE ACTION FOR ENTERPRISES DEPENDENT ON ADVANCED TECHNOLOGY

Summary

Companies can create a competitive advantage by combining high-quality human capital with sophisticated organizational capital. Four organizational types are based on these combinations: creative complexity, breakthrough science, applied science, and operational complexity. Five significant elements of knowledge capital creation capabilities are: know thy capital creation system, knowledgeable leadership, knowledge engineering, systems thinking, and continuous investment in systems of inquiry. This article provides guidance on leveraging the five elements to build knowledge capital for a target archetype.

Keywords: knowledge capital, human capital, organizational capital, competitive advantage, creating knowledge capital

CREATING KNOWLEDGE CAPITAL: AN ESSENTIAL COMPETITIVE ACTION FOR ENTERPRISES DEPENDENT ON ADVANCED TECHNOLOGY

1 Knowledge for Growth

The accumulation of knowledge is a key driver of sustained growth (Teece, 1998). Consequently, many nations and organizations strive to create a knowledge asymmetry, an amalgam of distinctive human and organizational capital (Figure 1). They strategically combine high-quality human capital with sophisticated organizational capital (Watson & Safadi, 2026). Traditional competitive advantages based on economic resources, physical assets, or market positions are quickly eroded by fast-changing markets. Knowledge-based advantages are more durable and harder to replicate (Cockburn et al., 2018).

Knowledge capital = Human capital + Organizational Capital

Figure 1: Knowledge capital is a composite form of capital

2 Knowledge Capital Asymmetry

Organizations increasingly deploy digitization to transform human capital into organizational capital, which is currently primarily institutionalized knowledge digitized as algorithms, automated procedures, and databases (Watson & Safadi, 2026). Machine learning and artificial intelligence threaten many professions, further exacerbating this conversion (Cockburn et al., 2018). Decision-making improves when information can be processed, allowing experts to reflect more on their knowledge and expertise (Simon, 1947). While organizational capital (e.g., a computerized accounting system) reduces the demand for low-level capabilities, it raises the need for highly skilled personnel to manage and exploit data analytics possibilities. Integrating

systems (e.g., an accounting system integrated into enterprise planning software) further ratchets up the demand for employees with high-level skills (Figure 2).

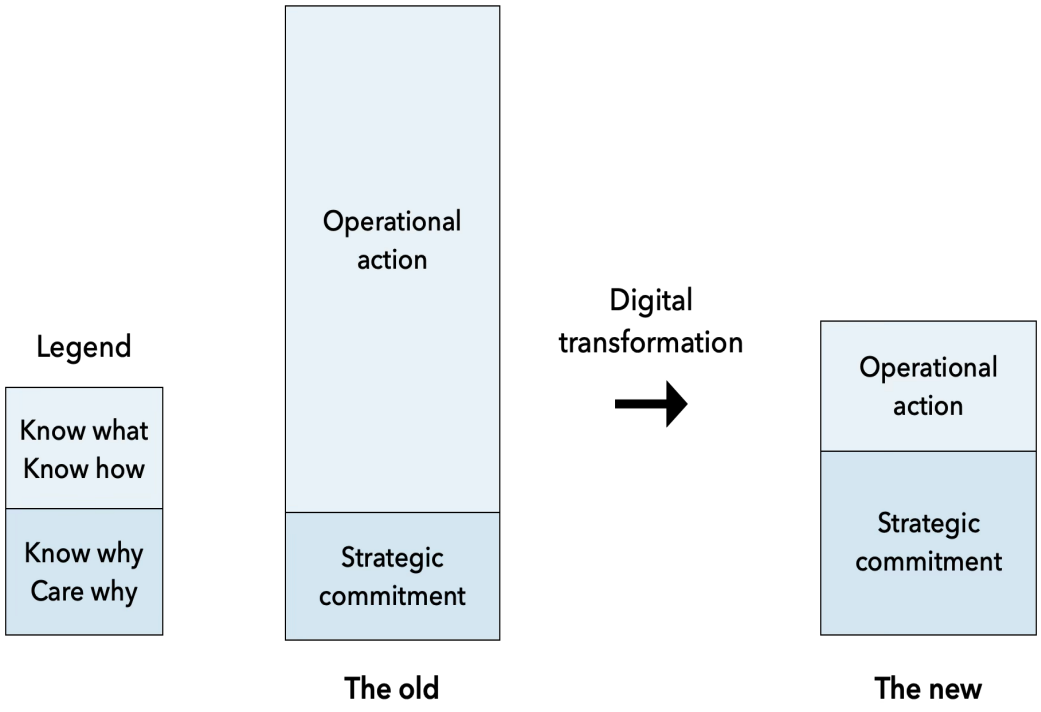


Figure 2: The effects of decades of digital transformation

Knowledge asymmetry as a means of competitive advantage was exploited by the Roman army, which had a manual of army procedures (organizational capital) and required legionaries (human capital) to possess basic literacy (Bernstein, 2013), while its foes were illiterate. A leading current example is Taiwan Semiconductor Manufacturing Company (TSMC). Although TSMC leverages machinery available to its competitors, its knowledge of configuring and operating semiconductor fabrication is hard to replicate. Asymmetry is not only about having smart people and superior systems but about the unique ways these two types of capital are reinforcingly integrated. Furthermore, its value is increased by domain expertise, social relationships that grease the cogs of change, and the ability to judge when and how to act.

The co-evolution of human and organizational capital (Figure 3) makes knowledge capital asymmetries sticky. Competitors need to replicate the human and organizational capital and their trajectory of parallel integration. It explains why semiconductor companies, such as Intel, continue to struggle to replicate TSMC's manufacturing performance despite investing significantly in human and organizational capital. Materializing a mutualistic symbiotic advantage between human and organizational capital is a significant organizational challenge.

An organization's intelligence, in order of increasing importance, is know what (cognitive knowledge), know how (advanced skills), know why (systems knowledge), and care why (self-motivated creativity) (Quinn et al., 1996). The lower forms of knowledge have been embedded in algorithms, and ownership has shifted from the individual (human capital) to the enterprise (organizational capital). Decades of digitization have (1) reduced the overall demand for human capital and (2) raised the demand for highly educated and dedicated personnel (Figure 3).

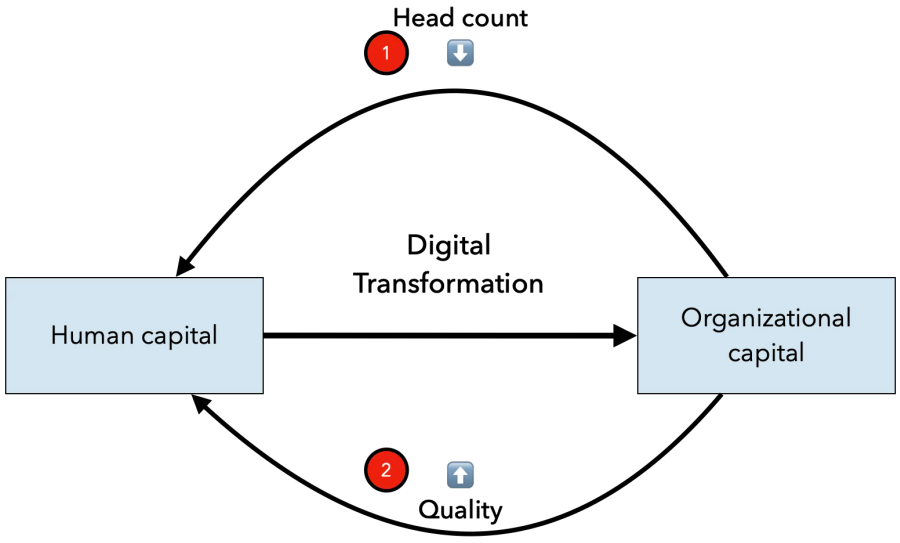


Figure 3. Digital transformation's impact on human and organizational capital

High-quality human capital understands how an organization creates value and has the wherewithal to raise efficiency and effectiveness to adapt to a changing environment. It is

typically highly motivated to lubricate the achievement of organizational goals. Briefly, it is the *know why* and *care why* elements (Quinn et al., 1996) that drive many organizations.

Intellectual and information processes create most of the value in an advanced economy. Geologists use visualization software to identify high-yielding deposits, and robots assist surgeons. Capital productivity grows when an organization can co-evolve human and organizational capital advancement. When people use decision-making software, they often identify new processes and data sources (organizational capital) that could further improve performance. There is a relentless pressure to capture internal and external data to create an integrated system of record that fuels a system of inquiry, such as machine learning, to extract insights and guide prediction (Watson, 2020).

Acquiring a knowledge capital asymmetry requires gaining a deep understanding of a domain and learning how to effectively integrate the relationship between human and organizational capital. Knowledge capital asymmetry also implies shifting the focus from means (such as technology) to ends (capital productivity).

3 Domains of Knowledge Capital Asymmetry

Capital can be combined in different proportions and to various degrees depending on the mix required to compete in an industry. It often implies a high-quality human capital (e.g., graduate degrees) leveraged by high-quality organizational capital (e.g., R&D labs and sophisticated algorithms) (Hausmann, 2016). We distinguish four combinations based on two levels of human and organizational capital, high and exceptionally high (Figure 4). Having identified the four combinations, we then searched for exemplars of each cell. We sought important, relevant recent examples, such as a COVID vaccine development and the Nobel prize-winning award for AlphaFold, and long-established examples of outstanding performance, such as Frank Gehry's

decades of architectural creativity and the US Navy's sustained deployment of the world's largest aircraft carrier fleet to exert global power.

		Human capital	
		High	Exceptionally high
Organizational capital	High	**Operational complexity** (An aircraft carrier)	**Applied science** (Vaccine development)
	Exceptionally high	**Breakthrough science** (AlphaFold)	**Creative complexity** (Design of an iconic building)

Figure 4. Combinations of human and organizational capital generate different forms of knowledge-driven organizations

3.1 Creativity Complexity

Renowned architect Frank Gehry had a well-defined design process, including the construction of wood and cardboard models (Flyvbjerg & Gardner, 2023). His warehouse of models (organizational capital) was a font of inspiration. After winning the Bilbao Guggenheim contract, Gehry spent two years iterating through analog and digital models. Dassault's software (organizational capital) for designing jets was modified for Gehry to represent and manipulate three-dimensional complexity. It amplified his creative genius by overcoming the difficulty of translating two-dimensional renderings into a form that builders can faithfully reproduce. However, as Gehry noted, "It's not just going to the computer, and creating a form, and then building a 3D model of it, and building it, and walking away" (Gehry, 2024). He acknowledged he also needs "this incredible staff that knows how to put things together and can pass muster in

the field in foreign countries". Gehry was a creative genius supported by a highly talented staff and highly specialized organizational capital.

3.2 Operational Complexity

A nuclear-powered aircraft carrier is an intricate operational platform. It can simultaneously face adverse weather conditions and sustained enemy assaults. US Navy carriers are operated by a young crew and a team of officers, with a 100% officer refresh cycle of about 40 months. Carriers develop and practice procedures (organizational capital) to handle dangerous situations. Nevertheless, a carrier's commanding officer observed, "A hundred things I have no control over could go wrong and wreck my career" (Rochlin et al., 1987). Successful operations depend on adaptable human capital with wisdom honed by decades of experience (Economist, 2025).

Three mechanisms build and maintain the high-quality human capital necessary to operate a carrier. First, a pool of senior chiefs (non-commissioned officers) rotates among carriers. They accumulate deep tacit knowledge from many years of practice of a specific skill. Second, many officers gain extensive experience from serving on multiple carriers. Third, continuity is maintained through ongoing rotation and personnel replacement (Rochlin et al., 1987). In effect, the US Navy builds a fleet of human capital that can quickly learn to command recruits effectively and dynamically tailor operational routines to changing circumstances.

3.3 Applied Science

BioNTech's scientists realized that they could apply their messenger RNA (mRNA) expertise to build a COVID-19 vaccine. By March 1, 2020, with around 20 vaccine candidates, it created a partnership with Pfizer, whose Chief Scientific Officer opined, "We believe that by pairing Pfizer's development, regulatory, and commercial capabilities (organizational capital) with

BioNTech's mRNA vaccine technology and expertise as one of the industry leaders, we are reinforcing our commitment to do everything we can do to combat this escalating pandemic, as quickly as possible" (Bourla, 2021). Pfizer had the organizational capital to accelerate vaccine approval, manufacturing, and global distribution. By December 2020, 74 million doses had been manufactured. The marriage of BioNTech's exceptional human capital and Pfizer's high-quality organizational capital was essential to delivering an approved vaccine rapidly on a global scale.

3.4 Breakthrough Science

Organizational capital is exceptional when it can exceed humans' collective creative capacity for specific complex tasks. For example, solving the "protein folding problem" has been a significant open research problem for over 50 years. The structure of about 100,000 proteins has been determined empirically, but there are billions of known protein structures. AlphaFold, machine learning software, can predict a protein's structure at near experimental accuracy in many cases (Jumper et al., 2021), and its developers won the Nobel Prize in Chemistry in 2024.

AlphaFold's development has a long pedigree. The more than 30 authors of the seminal paper drew upon the knowledge and data accumulated by prior scientists and on public-domain Python software. Science and software are accumulative organizational capital mechanisms that can reach a tipping point when integrated with high-quality human capital to create exceptional organizational capital, such as Alphafold. In turn, such exceptional capital provides a foundation for future breakthroughs, such as CRISPR for gene editing.

Science and software are emblematic of the capital creation systems humans have developed over thousands of years. Each generation accumulates capital that becomes the foundation of the capital created by the next generation (Watson, 2020). Recent generations have become remarkably effective at creating an educational system to build human capital and facilitate the

development of organizational capital in the form of procedures, databases, and software. The resulting knowledge capital is now a key determinant of success in many critical human endeavors and the operation of complex organizations.

4 Organizational Action to Create Knowledge Capital

A dynamic, changing environment necessitates that organizations create a knowledge advantage to prosper (Nonaka, 1994). The increasing complexity of a highly connected society requires executives skilled in knowledge co-creation and with experience with knowledge creation systems (Kreiling & Paunov, 2021). Consequently, firms need organizational structures, leadership programs, and incentives to build knowledge capital (Teece, 1998). Leadership programs are particularly required by multinationals operating in diverse regions and dependent on advanced technology. Our slate of organizational actions is aimed at such enterprises. While many smaller firms also need knowledge capital, they often lack the wherewithal to make extensive, sustained investments in executive development.

We identify five actions, one of which has three subsections, necessary to pave the way to knowledge asymmetry. Before contemplating action, executives might find it fruitful to administer a diagnostic survey to assess whether any of the recommendations are currently active and to what extent.

4.1 Know Thy Capital Creation System

There must be a collective awareness of organizational goals and the business logic that guides capital creation. Understanding one's capital creation system[1] helps an organization locate itself within the knowledge asymmetry quadrant (Figure 4). While it can be difficult to specify (high or very high) the levels of human and organizational capital, an organization can assess itself relative to competitors and set aspirational levels of capital. What levels of capital does competitive advantage require? Does it achieve these internally or via a partnership?

4.2 Job Rotation to Create Knowledgeable Future Leaders

A knowledge asymmetry is based on recruiting graduates with the required general knowledge, a yen for learning, and an aptitude for leadership. To be effective leaders, they need domain knowledge, judgment skills, and social capital, which can be built through programs such as job rotation, practiced in Japanese firms since at least the 1950s (Koike, 1984).

Job rotation is a structured mechanism for building a cadre of future leaders skilled in knowledge transfer (Soo et al., 2002).It must include activities that deploy introspection and dialogue to mold mental models germinated by operational experience and exposure to high-level decision-making. Similarly, the growth of social capital must be deliberately planned and fostered.

Innovation increases when employees transfer knowledge laterally from one task to another (Coşgel & Miceli, 1999). Firms applying job rotation are better able to manage change (Koike, 1984). Leadership development is more effective when it primarily rotates high performers

[1] For a method depicting a capital creation system, see Figure 4 in Watson, R. T., & Safadi, H. (2026). Capital asymmetry: A lens for strategic analysis. *Business Horizons*, *69*, 395–405.

(Kampkötter et al., 2016). Rotation of future executives can build their domain knowledge, judgmental skills, and social capital.

4.2.1 Domain knowledge

Domain knowledge (Alexander, 1992) includes the know what (cognitive knowledge), know how (advanced skills), and know why (systems knowledge). It contributes to operational success, particularly when handling atypical situations (Aoki, 1986). Diverse operational experience enables future leaders to gain in-depth familiarity with an organization's key capital creation systems and learn how to use them to co-create knowledge with internal and external partners. They should gain managerial experience that enlightens them on how vital functions connect to create the whole enterprise. As appropriate, they should also have operational management experience in different societies to become sensitized to the impact of culture on behavior and decision-making (Hofstede, 1980).

4.2.2 Social capital

Social capital is a network of connections, shared values, culture, and context that support social cohesion, trust, and teamwork (Watson, 2020). It promotes knowledge acquisition (Soo et al., 2002) and *care why* by nurturing organizational attachment (Festinger et al., 1963). Social capital enables a leader to achieve their agenda by drawing upon established friendships, mutual trust, and organizational affinity to support decisions and actions. Because humans have a strong desire to bond (Lawrence & Nohria, 2002), organizations can build social capital by structuring peer and superior interactions to foster the growth of a future leader's social capital.

4.2.3 Judgment skills

Good judgment is a critical leadership skill because decision-making is the central activity of an organization (Huber, 1984).It typically requires tacit knowledge (Tung & Roussiere, 2026), a

type of non-codifiable knowledge acquired informally through diverse encounters that formulate generalizable mental models for decision-making (Johnson-Laird, 1983). These models support frameworks that enable individuals to perceive the world and envision the future. Judgmental skills, like tacit knowledge, can be developed through socialization and internalization (Nonaka, 1994). Socialization involves observation, imitation, and practice. For leadership development, this means potential executives must serve an apprenticeship in critical deliberative reasoning (Mercier & Sperber, 2017), allowing them to observe the nuances and participate in the associated social interactions. Internalization describes the conversion of domain knowledge into mental models. Potential leaders need to learn to transform their experiences into frameworks, models, and theories. Judgmental skills are essential for strategic decision-making, deciding with whom to co-create capital and knowledge, and imagining an organization's future.

4.3 Knowledge Engineering

Knowledge engineering is the design and maintenance of a system of inquiry. It sustains a dynamic entanglement between human and organizational capital to ensure that knowledge maintains action-relevant meaning (Fahey & Prusak, 1998). We advocate deploying knowledge engineers to monitor the link between human and organizational capital. They are responsible for learning how a domain specialist's performance could be improved by adding features to supporting systems and communicating their evaluation to the information systems team. Their role is bidirectional, so they are also accountable for educating human capital in the productive use of organizational capital. Knowledge engineers are multi-skilled boundary-spanning communicators. Investments in information systems and AI must be complemented by the development of human capital that can effectively exploit new information processing

capabilities, and, in turn, guide future IS and AI investments towards higher-level individual, team, and organizational productivity.

As AI becomes the dominant methodology for developing systems of inquiry, the significance of knowledge engineering becomes increasingly apparent. Knowledge engineers play a crucial role in categorizing and managing data, enabling AI systems to comprehend specific domains. A notable application of knowledge engineering is retrieval-augmented generation, which blends knowledge graphs and generative AI to find patterns in enterprise data (Bratanic & Hane, 2025). Knowledge engineers design systems that leverage domain knowledge to facilitate deliberative decision-making.

4.4 Systems Thinking

When addressing the US Congress, Jay Forrester asserted, "All decisions are made based on models. All laws are passed on the basis of models. All executive actions are taken on the basis of models" (Forrester, 1971). He advocated the use of systems dynamics for addressing complex, highly interactive, nonlinear systems. Systems thinking skills can be built as part of developing future leaders' judgment skills by requiring them to translate their experience into decision-making models and frameworks. Decision-makers, and especially knowledge engineers, need skills in systems dynamics to comprehend the internal and external consequences of their actions.

In parallel, executives must develop a comprehensive understanding of a firm's key operational systems. We identify five fundamental systems necessary for managing an organization (Table 1). These systems should be interconnected to enable an organization to extract maximum value from its data resources, thereby enhancing its capital productivity (Figure 5).

Table 1: Types of systems (Watson, 2020)

Type	Purpose
Engagement	Collaborate and coordinate activities, such as meetings and plans.
Framing	Justify the reason for behaving in a particular way or promoting a particular opinion—a company's value proposition.
Inquiry	Generate knowledge, such as data analytics and machine learning.
Production	Create and transport products and services.
Record	Record and retrieve data, such as a database or document store.

An enterprise has a variety of *systems of engagement* to woo customers, coordinate internal activities, and influence major stakeholders. It has *a system of production* to deliver products and services. A *system of record* stores engagement and production data, which its *system of inquiry mines* to feed knowledge into a *system of framing* that identifies potential improvements and refinements to an organization's value proposition.

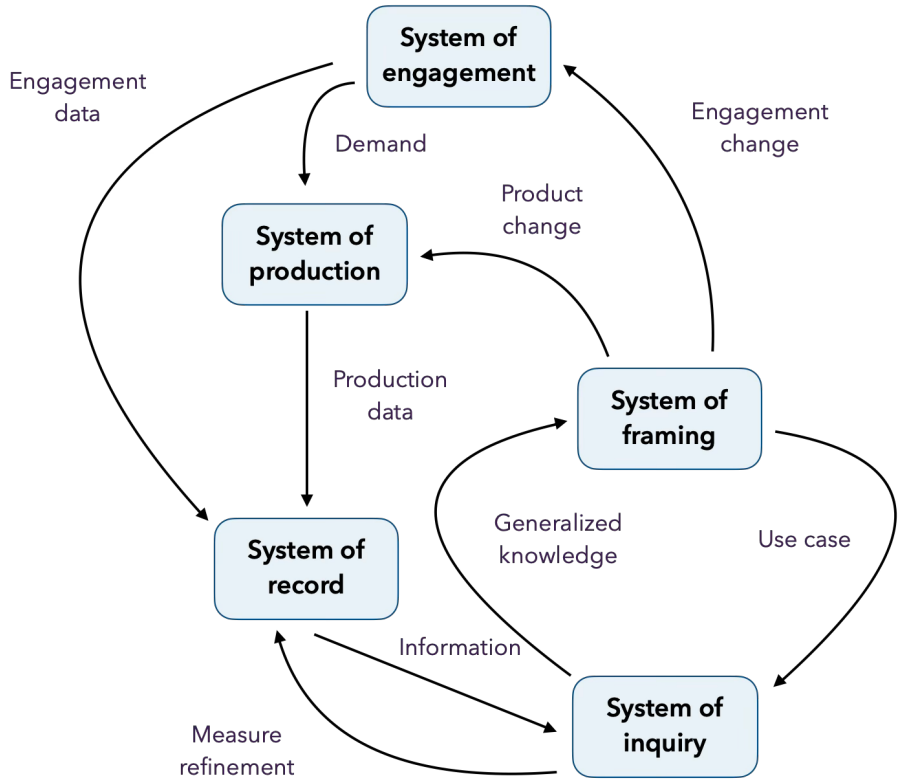


Figure 5: A virtuous systems cycle

4.5 Continuous Investment in a System of Inquiry

Because knowledge deteriorates (Fahey & Prusak, 1998), a system of inquiry is central to improving capital productivity, serving as a framework for continually acquiring and processing knowledge (Churchman, 1971). It creates new generalized knowledge by processing emerging data to identify patterns. A system of inquiry reinforces learning when decision-makers identify use cases they want analyzed, which may precipitate the collection of additional data. The potential value of a system of inquiry has increased with advances in data processing, including query languages, data analytics, and machine learning. Data-driven organizations are more successful (McAfee & Brynjolfsson, 2012), but only if they adapt their deliberations to incorporate advances in extracting value from data. Simultaneously, they need to invest in data capture to gain a deeper understanding of their stakeholders and operations.

5 Customizing Knowledge Creation to an Organization's Archetype

The four archetypes (Figure 4) require different capabilities, built on a shared foundation of domain knowledge and social capital.

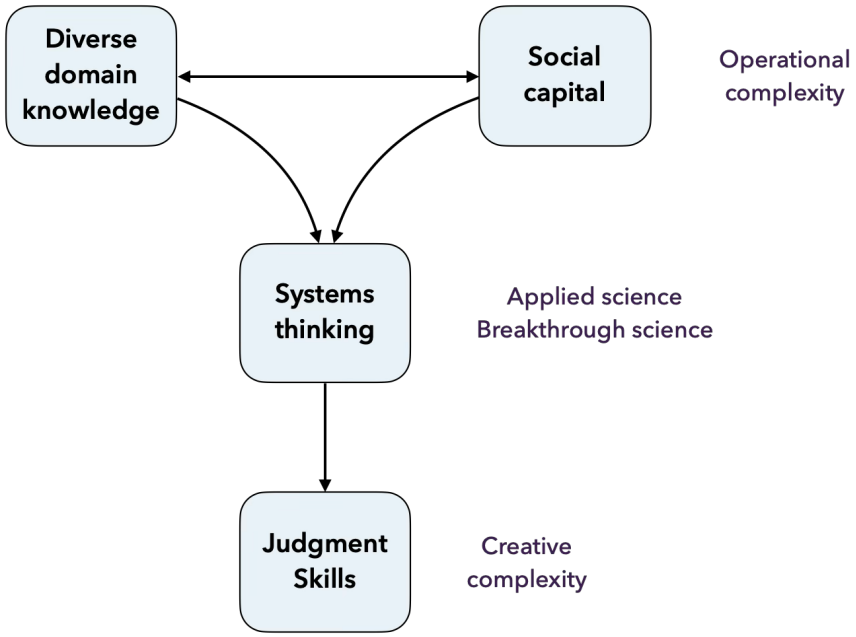


Figure 6: Knowledge capital creation goals for knowledge-driven organizations

5.1 Operational Complexity

Close-knit, skillfully diverse functional teams are an effective way to quickly address operational complexity. Playbooks and standard procedures exist, yet decision-makers can adapt them on the spot because of their extensive experience. An established social network helps communicate information and share expertise efficiently.

5.2 Science

Applied scientists deploy what is known to solve problems. Given the complexity of many issues, there is a need for diverse domain knowledge. Boundary spanners can be particularly valuable because they can contribute different levels of knowledge, expertise, and research

methodologies (Goodrich et al., 2020). Translating science into products requires end-to-end processes and supply chains (e.g., BioNTech and Pfizer collaboration). Social capital enables speed but is not the differentiator. Systems thinking is needed to ensure continuity between the various types of systems (Figure 5).

Breakthrough science requires a deep understanding of one or more fields (e.g., materials science and quantum mechanics), the ability to see how parts contribute to an innovative whole, and the judgment to identify which of these wholes offers the most potential for a breakthrough. Innovative scientists need a vision of what could be learned if they follow a carefully chosen line of discovery. Systematic recombinant innovation depends on an architecture for integrating knowledge rather than on lone-genius discoveries. Social capital enables collaboration across boundaries, and systems thinking facilitates the assembly of impactful solutions.

5.3 Creative Complexity

Creative complexity requires an exploratory process in which multiple alternatives are surfaced, critically reviewed, and refined – and possibly discarded. There is usually an innovative leader (or a small, long-intact team), who sets the agenda and reacts to critiques. Ultimately, the leader concludes what is feasible and sets the future direction. While a leader's judgment can be contentious, it is typically respected due to their past successes and deep social capital.

6 Conclusion

Knowledge has always been a strategic factor for innovation, productivity growth, and organizational success (Sayyadi & Provitera, 2022). Technology affords new forms of organizational capital that can be mined by appropriately skilled people for operational action and decision-making. Digital technology has strengthened the bond between human and

organizational capital, further cementing knowledge capital as the mainstay of success – irrespective of form, size, or purpose.

The impact of AI on organizational knowledge creation is unresolved. LLMs can accelerate knowledge retrieval and sharing. However, there is the danger that human creativity will atrophy (Chin et al., 2026). Knowledge summarization eliminates the benefits of deep reading, such as serendipitous insights and vicarious learning (Bandura, 1977). Writing refines thinking as the author has to struggle to explain ideas to themselves before attempting to influence others. In a knowledge-dependent society, shortcuts might be long detours to less knowledge. AI will not dissipate the advantages of a capital asymmetry that is deeply embedded in an organization's thinking, culture, and executive development.

When knowledge capital asymmetry becomes a dominant form of competitive advantage, it can create monopolies, such as Amazon in retail (Safadi & Watson, 2023). Because knowledge capital is unevenly distributed, a small group of exceptional human capital can emerge, with major tech massively rewarding a few AI scientists while laying off tens of thousands of employees. This is a critical, persistent effect of a knowledge-centric economy that widens inequities and profoundly impacts social cohesion.

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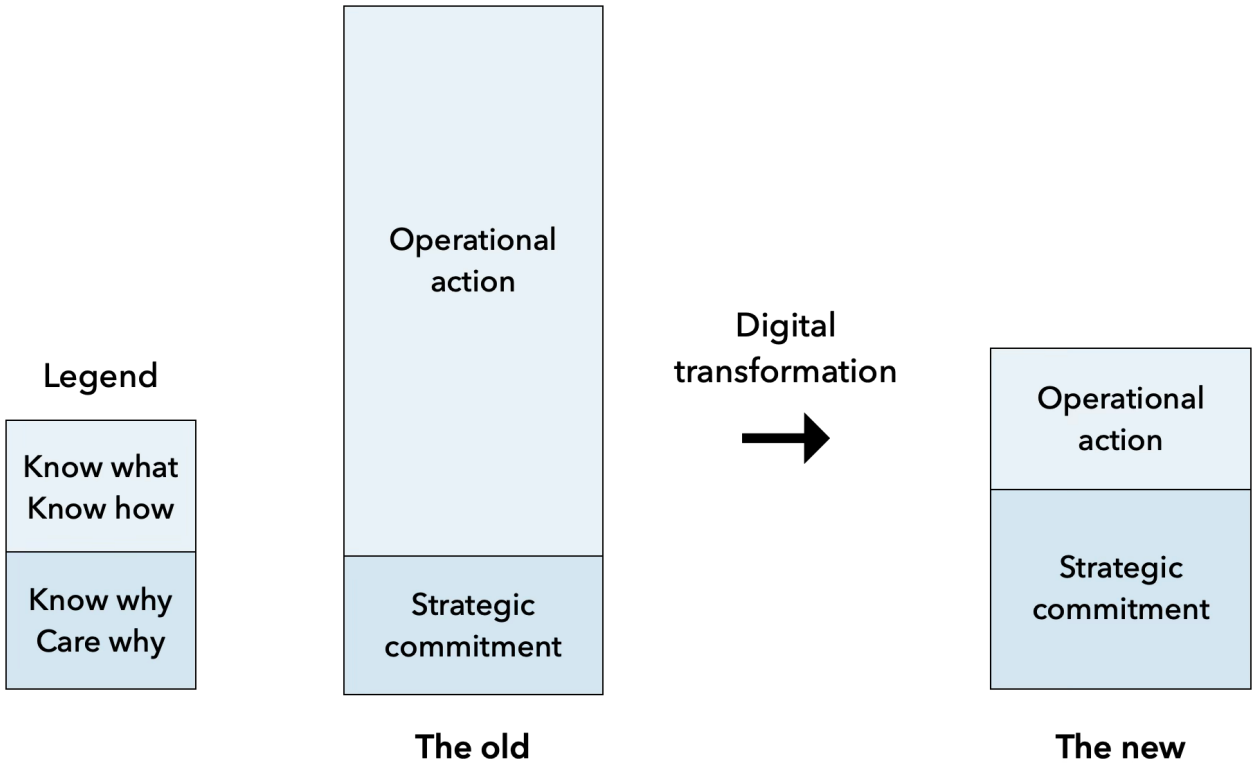


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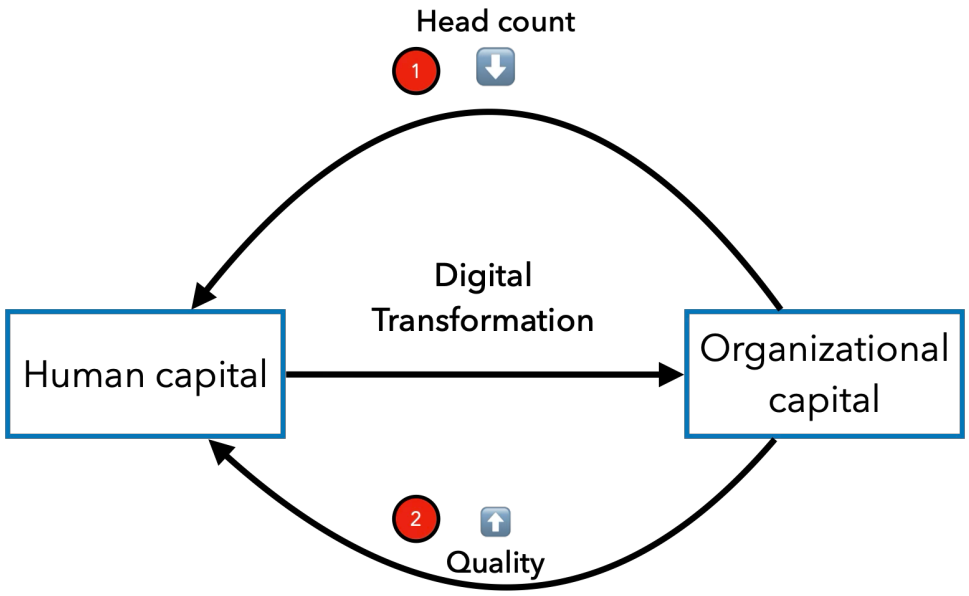


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Inquiry	Generate knowledge, such as data analytics and machine learning.
Production	Create and transport products and services.
Record	Record and retrieve data, such as a database or document store.

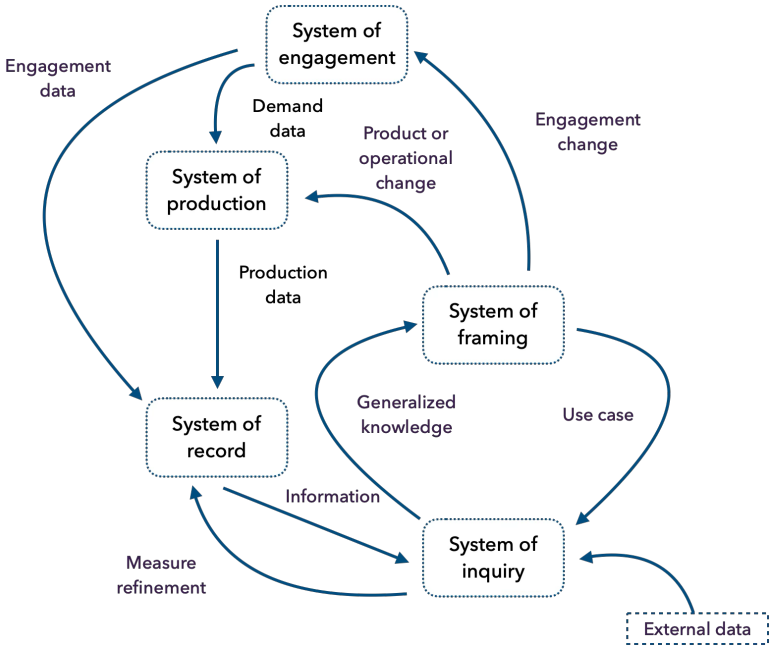


Figure 7: A virtuous systems cycle

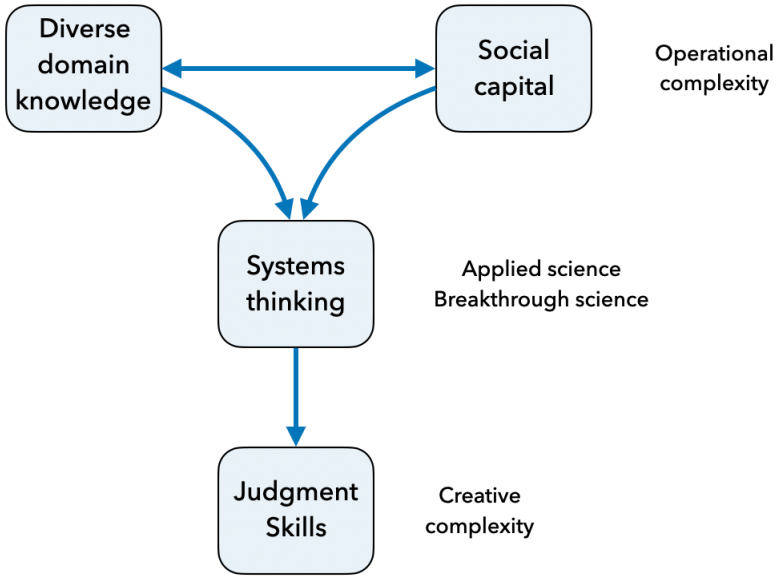


Figure 8: Knowledge capital creation goals for knowledge-driven organizations